Economics of Health and Health Care

Lecture 9: Production of health care

Pedro Pita Barros
Universidade Nova de Lisboa

Production in health care providers

- What is dfifferent in a hospital?
- Meaning of efficiency
  - Technological
  - Allocative
  - Economic
- Three questions regarding technology
  - Degree of substitution between inputs
  - Optimal size - scale economies
  - Joint products - scope economies
- Quick recall of basic concepts – example of hospital (there are other providers, of course)

What is a hospital? Ask AI

- “provides medical treatment through professionals and equipment”
- Expand to: “teaching, training health professionals, and conducting clinical research”
- “From an institutional perspective, hospitals can be public, private, or non-profit, and their objectives may vary.”
- “Economically, it acts as a complex firm combining capital and specialized labor to produce intensive inpatient and outpatient health services”
- “hospitals are (…) treated as multi-product firms operating in imperfectly competitive, highly regulated markets, where information asymmetries and externalities are central features"

2 In economic analysis, can we treat a hospital in an analogous way to common firms?

(select all that apply)

Response	Average	Total
yes, because hospitals use and have to combine inputs (equipment, different types of health professionals, etc)	89%	16
no, because hospitals care about patients and not about profits	11%	2
yes, because hospitals have administration boards and decide as firms	61%	11
no, because hospitals deal with life-threatening cases	17%	3

What makes a hospital distinct from a firm?

- Complex internal structure
  - Medical chain of command
  - Administrative structure
- Internal power relations are quite different from what you see inside companies
- Additional complexity comes from the not-for-profit nature of most hospitals

Hospital

- There are several questions to define the hospital as an economic agent:
  - Who has the rights of control?
  - Who decides within the hospital?
- Several possible answers
  - Physicians - they influence decisions in the direction they want
  - Hospital administrators - with their own objectives
  - The “owner” of the hospital, of whom the administration is a perfect agent

What is a hospital? A summary

- Admitting people to stay
- Double hierarchy - clinical and administration
- Several agents with decision power - owners, managers, health professionals (doctors)

Efficiency concepts

Avoid waste - technological efficiency - get to the production function frontier

Get the right combination of inputs - allocative efficiency - use input prices to define the best point in the production frontier

Produce the right amount - economic efficiency - get to the point of marginal benefit (price, in many cases) equal to marginal cost of production

More standard issues: technology

- Substitution between inputs
- But professionals think about fixed coefficients
- Estimates for the elasticity of substitution
- (General line: efficiency analysis – several methodological approaches exist, most popular are DEA and stochastic Frontier estimation)

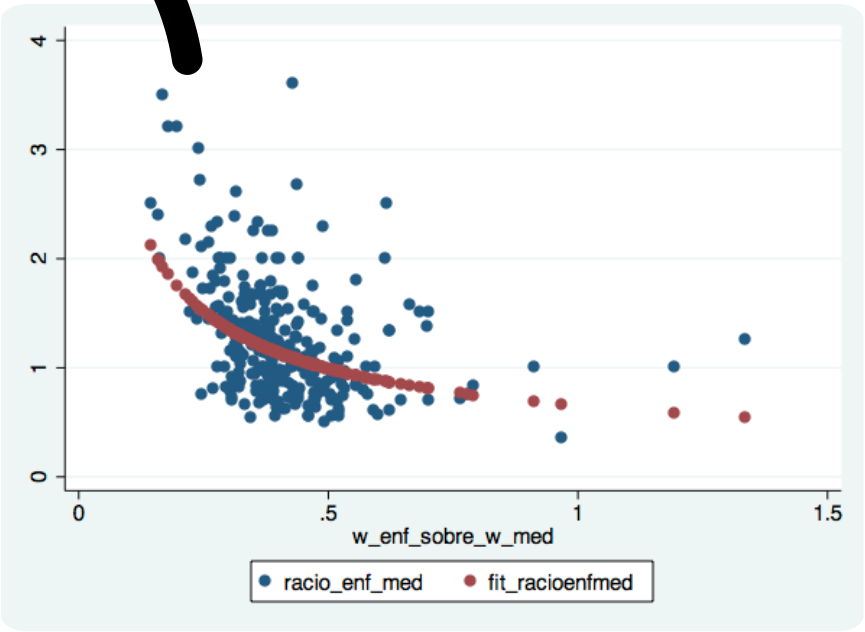
Elasticity of substitution

- Definition: how much the ratio of input factors change in % when the relative price of inputs change 1%

- It is related to the shape of the isoquant

Hospital =>	**Without teaching**	**With teaching**
Doctors vs. nurses	0,547	0,159
Doctors vs. beds	0,175	0,155
Nurses vs. beds	0,124	0,211
Nurses vs. interns		2,127
Doctors vs. interns		0,292

- Racio doctors/nurses vs relative wage – seems to have a negative relation as predicted by substitution of inputs



Technology: economies of scale

- Definition of **economies of scale**: An increase in production reduces the average cost
- Formal definition: $AC(Q) = C(Q)/Q$, where $C(Q)$ is the cost function, and $dAC/dQ < 0$
- Counterpart of increasing returns to scale in the production function
- Implication: efficient units of different sizs will have different average costs

Waiting lists and waiting times

3 The existence of waiting lists and positive waiting times is a symptom of health system inefficiencies.

Response	Average	Total
Yes	24%	4
No	76%	13

Waiting lists: Some theory

A simple example

- Patients arrive every 30 min
- Treatment takes either 15 or 45 min, with prob 1/2
- What's the average treatment time?
  - 30 min, of course
- What's the average waiting time?
  - It will not be zero...
  - Take the second consultation, what is the expected time?
  - First one took 15 min, waiting time is zero with prob 1/2
  - First one took 45 min, waiting time is 15 min with prob 1/2
  - Average waiting time is 7,5 min
  - Do it for the third one - again 7,5 min

What is the result?

- Note that not only we have waiting lists, we also have idle times!!
- We can generalize this result, though the same intuition holds
- On social terms, having zero waiting time is probably too costly
- Strategic behaviour of agents

Public vs private sector

- Waiting lists are often seen as bad per se
- Private sector activity is seen as a way out, a solution to this problem
- Iversen (1997) looks at this - patient has a trade-off between zero cost in public, and price *p* in private

Public vs private sector

- Public sector may “game” on this - increase waiting time in public sector to lead to exit to private sector; smaller public sector, cost savings
- Thus, waiting lists and waiting time may result from strategic action from the public sector concerned with costs

What can we expect from agents' choices?

- Another concern with waiting lists:
  - Private sector only attempts to attract good cases
  - Complex cases remain in the public waiting list
  - There is cream skimming
- This is not necessarily the case
  - The "common wisdom" argument ignores that patients have a say - different patients have a different waiting time - price trade-off
  - The ones with least severe conditions will remain in the public, even if the doctors try to cream-skim
  - Average complexity in public sector may be higher or lower than in public sector

What can we expect from agents' choices?

- We can discuss under what conditions cream-skim can exist
- And show that in general the public system will serve the extremes of the market - the ones the private sector does not want; the ones that do not want to leave the public sector because they have a low cost of waiting
- Average case-mix in public can be higher or lower than in private sector

General approach: Where to look for waiting times?

- 1: visit to primary care center – **wait** for appointment
- 2: referral to hospital: **wait** – from request to put in the schedule; from it to the day of the appointment at hospital
- 3: indication for surgery: **wait** – until it is scheduled, after scheduled until day of surgery => most discussions focus on this

Waiting lists and waiting times

May result from simple randomness of arrivals of patients and treatment times

It is another “tool” to balance demand and supply

May result from deliberate decisions of economic agents (hospitals)